

Your grade: 100%

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Next item →

1. Consider a system having nonlinear state equation

1 / 1 point

$$x_k = 3x_{k-1}^2 + 2u_{k-1} + \sqrt{w_{k-1}}.$$

If $\hat{x}_{k-1}^+ = 1$, $u_{k-1} = -1$, and $\bar{w}_{k-1} = 4$, what is $\hat{A}_{k-1}$? Enter your answer with one digit to the right of the decimal point.

6

✔ Correct
Yes. That is the correct answer.

2. Consider a system having nonlinear state equation

1 / 1 point

$$x_k = 3x_{k-1}^2 + 2u_{k-1} + \sqrt{w_{k-1}}.$$

If $\hat{x}_{k-1}^+ = 1$, $u_{k-1} = -1$, and $\bar{w}_{k-1} = 4$, what is $\hat{B}_{k-1}$? Enter your answer with two digits to the right of the decimal point.

0.25

✔ Correct
Yes. That is the correct answer.

3. Consider a system having nonlinear measurement equation

1 / 1 point

$$z_k = \sqrt{x_k + u_k} + v_k.$$

If $\hat{x}_k^- = 16$, $u_k = 4$, and $\bar{v}_k = 0$, what is the value $\hat{z}_k$ computed by an extended Kalman filter? Enter your answer with one digit to the right of the decimal point.

8

✔ Correct
Yes. That is the correct answer.